SafeKid Scan: Early Detection of Digital Addiction in Minors

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***Abstract*—The lack of awareness of the consequences of the rapid development of social media use among children and adolescents has increased the risk of digital addiction, impacting academic performance, behavior, and mental health. Current detection tools mainly rely on surveys or screen-time data, which are subjective and unsuitable for real-time intervention. To overcome these limitations, this paper presents a smart and privacy-enhancing architecture to early identify and manage social media addiction in minors aged 10–18. The system combines multimodal image, caption, and hashtag processing in English and Sinhala to generate addiction risk scores. It also integrates complaint-based risk detection, AI-driven decision support, blockchain-based information integrity, and IoT-based behavioral monitoring. Experimental evaluation achieved 95.97% validation accuracy for media analysis, 85.56% for complaint analysis, and 96.97% for IoT-based monitoring. The framework supports accurate risk identification, timely intervention, and safer digital environments for children and adolescents.**

Index Terms*—*Digital Addiction, Social Media Analysis, Multimodal Image Processing, AI-Based Risk Detection, IoT Monitoring

I. Introduction

The use of the social media among the children and adolescents is a pervasive phenomenon. Social networking web sites and content sharing applications are widely used in communication, entertainment and to carry out education. Although these technologies have numerous advantages, one significant issue has appeared out of the uncontrolled, even excessive, use of them that can result in the creation of digital addiction. Especially vulnerable is such a demographic group of individuals between the age of 10 and 18 because the age-specific period is critical in terms of the establishment of emotional stability, decision-making abilities, and social growth[1]. In the recent empirical reports, it was stated that constant social media use can be harmful to the academic performance of young people and their personal well-being. The most common negative effects are mentioned in the form of low concentration, sleep problems, increased anxiety, social isolation and emotional volatility. The visual appeal, emotional words, or viral posts, demanding repetitive interaction, are often the triggers of these issues. These stimuli lead to slow development of addictive habits which may not be noticed during the early stages.

Currently, the majority of diagnostic methods of digital addiction depend on the questionnaires, self-reported screen-time data or manual confirmation by parents and educators. Although these methods bring simple observational information, the real-time validity and practicality is less in them. Moreover, modern systems tend to concentrate on one aspect, either one in the form of textual analysis or the duration of usage, yet social media content is usually a combination of images, captions, hashtags, and interaction patterns which affect the user behavior as a whole. Some of the shortcomings of solutions used are related to inability to adapt to the fast changing online environment. The trend of social media changes very quickly and makes strict systems outdated after a certain period of time. Also, too little focus is put on multilingual contexts; in South Asian areas, the users regularly switch between a native language and English in the same text, which complicates the application of a current analytical model[2]. In an attempt to mitigate these shortcomings, this paper suggests an intelligent surveillance and intervention model that will identify digital addiction among minors that can be attributed to the exposure of social media. The system analyses the behavioral information to identify online interaction patterns in liaising with guardians and authorized supervisors. All data are automatically processed

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within a secure framework that prioritizes privacy and controlled access. Additionally, the platform incorporates optional Internet of Things (IoT) devices that can be deployed when required. These devices enable continuous behavioral and physiological monitoring, providing psychiatrists and mental-health professionals with objective insights for more informed clinical decisions. By integrating automated multimodal analysis, secure data handling, institutional collaboration, and clinically oriented support, the proposed system addresses the limitation of single-modality, non-real-time detection approaches, offering a comprehensive and reliable mechanism for safeguarding minors in digital environments.

The paper is divided into five parts. The section II is a review of the literature and background research available regarding the topic of digital addiction detection and multimodal analysis. In Section III, the architecture and methodological framework of the proposed system of the SafeKid Integrity of Knowledge: SafeKid Scan, including the multimodal media analysis, a complaint-based risk assessment, and an IoT-related monitoring, are outlined. Section IV will provide the results of the experiment and the analysis of the efficiency of the suggested framework. Lastly, the last Section gives final remarks and outlines the future research directions.

II. LITERATURE REVIEW

The blistering development of digital devices and the popularity of social networking sites have changed the contemporary lifestyle to a great extent. Facebook, Instagram, Tik Tok, YouTube and instant messaging service applications are now a fundamental part of communication, sharing information, and entertainment. Although these platforms can be of great benefit, there has been too much uncontrolled consumption of these platforms leading to the rise of digital addiction as an important behavioral and psychological issue. Pioneering clinical studies classified internet addiction as a specific disorder, which is marked by the obsessional use of the internet and the inability to control [1]. Later research also indicated that social network sites enhance addictive tendencies since they promote unceasing social validation and interaction processes [2]. The early studies on digital addiction were majorly psychological and behavioral in nature. Researchers studied the problem of social media dependency using self-report surveys, interviews, and addictive behavior inventory and found such symptoms as tolerance, withdrawal, compulsive checking, and mood alteration [3]. These conventional assessment methods were significant in creating theoretical bases of learning addictive behaviors. Nevertheless, subsequent research revealed some fundamental flaws of these methods, such as subjectivity, poor self-awareness and insufficient potential of real-time observation of behavior [4].

With the emergence of the digital platforms that started producing large volumes of user information, the approach to research changed toward data-driven and computational approaches. The machine learning and automated analytics enabled the identification of behavioral patterns in large-scale data sets on the Internet, and multimodal deep-learning strategies have shown an ability to recognize intricate behavioral cues with online communication more effectively than human evaluations [5]. Such calculations significantly improved scalability and objectivity, which allowed early detection of mental health risks associated with overuse of digital platforms. Subsequent progress saw the introduction of machine learning designs that were able to analyze social network data in order to identify mental illnesses like depression and problematic usage behavior. Decision trees, support vector machines, and ensemble -based algorithms were observed to be highly predictive when used with the data related to user interaction, frequency of posting, and metrics of engagement [6]. These results supported the validity of machine learning algorithms to detect the behavioral anomalies related to digital addiction.As the dominance of content-centered platforms grew, the studies started to concentrate not on the usage statistics but on content-based ones. The sentiment analysis and opinion mining approaches allowed the expression of emotional polarity and subjective utterances to be extracted out of user-generated text in the form of posts, captions, and comments [7]. Such strategies enabled scientists to associate such patterns of negative sentiment, emotional instability, and obsessive use of language with problematic digital behavior. Deep-learning methods also enhanced the state of textual analysis by efficiently encoding the contextual and sequential dependencies in the natural language. Neural, convolutional, and recurrent networks have demonstrated superior performance in sentence-level emotion and text recognition tasks, particularly when applied to large-scale social media datasets [8]. The methodologies, therefore, contribute to the accuracy of deriving psychological-distress indicators of continuous textual data streams.

At the same time, there has been an increase in the popularity of visual-based analytics due to a growing popularity of images on social media. Deep convolutional neural networks enable automated retrieval of fine visual features the expressions on faces, feelings of affection or anger, groups of colors, and repetitive patterns. Advanced studies in image deep classification have revealed that Convolutional Neural Networ(CNN) structures are capable of identifying high-level visual image features with exceptional precision [9]. Such analysis as visual-content analysis can be especially useful, as visual images often represent the affective states, which cannot be expressed in words by the users. The recent scholarship underlines the importance of multimodal machine-learning models that integrate both textual, visual and behavioral data. Complementary signals are used by multimodal systems as features at the feature-level and as decision-level to exploit disparate data modalities. It has been shown empirically that these frameworks generate better results on human-behavioral analysis than do unimodal approaches because they capture complex cross-modal relationships not reachable by single-modality models [10]. Thus, multimodal learning will be a key innovation in the area of digital-addiction detection. The study explored speech-based emotion recognition

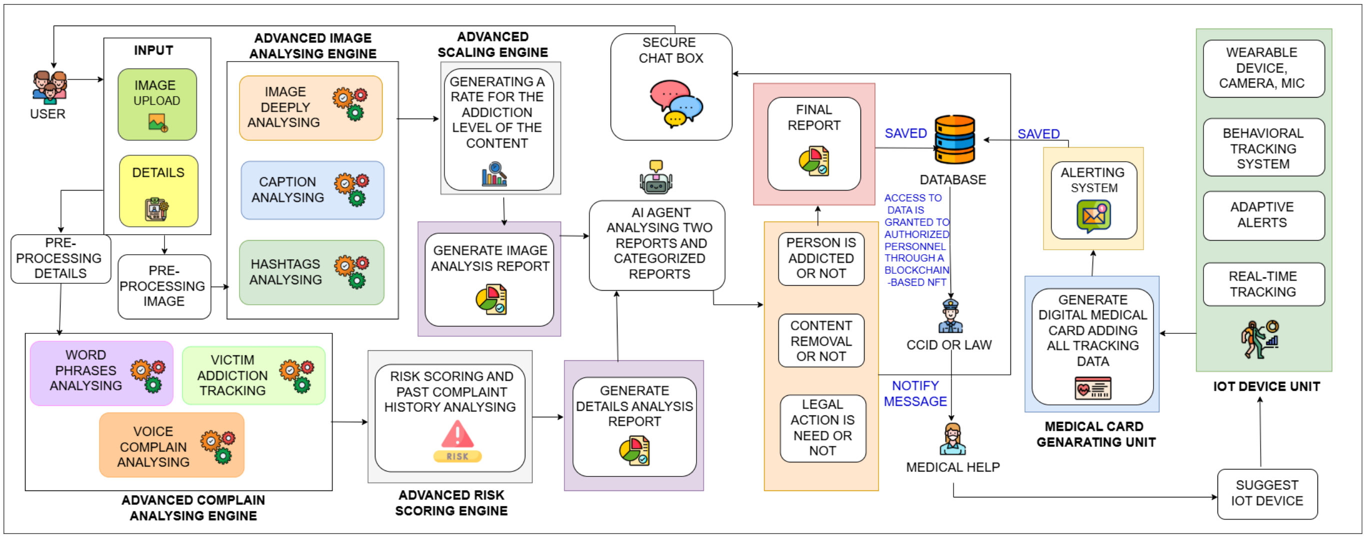

Fig. 1: System architecture diagram

Emotional distress and compulsive behavior can be further analyzed using acoustic features, the variation of pitch, energy allocation, tonal quality, and speech rhythm extracted through the speech processing model. The development of advanced adversarial and semi-supervised learning methods has demonstrated encouraging outcomes in the speech emotion recognition task and significantly improves the robustness when using it in combination with other modalities [11]. With these immense technological developments, there are still a number of challenges that have not been addressed. The major weakness lies in lack of system level integration. Some of the studies focus on individual elements of analysis without creating end-to-end pipelines that can process information to include data acquisition, preprocessing, feature extraction, prediction, and reporting in a single framework. Besides, ethical and privacy issues are often under-considered. The requirement of mental health data is important to protect privacy, informed consent, and safe data processing tools to maintain responsible deployment [12]. Altogether, available literature supports the usefulness of machine learning, deep learning, sentiment analysis, image processing, and multimodal learning solutions in recognizing digital addiction and other associated psychological risks[12]. However, the limitations to the practical deployment are fragmented system design, lack of longitudinal analysis, and insufficient ethical protection. Such constraints emphasize the need to have an integrated, privacy conscious, multimodal digital addiction detection system that can be used to support real world interventions and may be used to inform early intervention initiatives.

III. METHODOLOGY

This part outlines the methodology that will be used in the framework of the SafeKid Scan to identify the latent digital addiction tendencies among the minors. As shown in Figure 1, The system uses a unified pipeline combining AI-based multimodal analysis and IoT behavioral monitoring to assess visual, textual, and physiological risk signals. Social media and complaint data are analyzed first, followed by continuous IoT monitoring under privacy-aware guardian supervision to enhance detection accuracy.

A. Image-Based Media Pattern Analysis

This module analyzes multimodal social media content shared by minors to detect early indicators of addictive or harmful behavior. Each post comprises an image, caption, and hashtags, with images preprocessed through resizing and normalization to ensure consistency. To achieve contextual understanding of media content, a contrastive language–image model similar to Contrastive Language-Image Pre-Training(CLIP) is used to encode images and textual metadata into a shared semantic embedding space. Visual features are extracted using a deep convolutional neural network based on the ResNet architecture, which is widely recognized for its effectiveness in deep image representation learning. Caption and hashtag text are processed using lightweight natural language preprocessing techniques, including tokenization, sentiment estimation, and keyword-based relevance scoring.

The training dataset consists of 34,900 images grouped into three categories of addiction-related content commonly observed in Asian social media environments: Explicit harmful addiction content, such as illegal drugs, weapons, and alcohol; Visual addiction content, including video games, anime, and meme-based media; and Psychological trigger content, such as graphic violence, fights, and aggressive behavior. The images were collected from the Sri Lanka Child Care Authority, publicly available social media platforms, and curated Asian-region digital content repositories. These categories enable the model to learn diverse visual and semantic patterns associated with risky or addictive online exposure. During training, multimodal features are processed by a supervised classifier that maps media representations to continuous addiction risk levels rather than binary labels, allowing for more fine-grained and interpretable assessment. To quantitatively assess the risk associated with each social media post, the proposed system computes a media risk score using a Weighted Linear Combination (WLC) strategy. Three complementary indicators are considered in this formulation. The image-based risk score, denoted as S_{image}, is derived from high-level visual features extracted through convolutional neural network (CNN) analysis. The caption-based textual risk score, represented as S_{capt}, is obtained by applying sentiment analysis and keyword relevance estimation to post captions. In addition, the hashtag-based risk score, S_{hash}, captures semantic trends and contextual cues reflected through hashtag usage. These indicators collectively contribute to a unified and interpretable media-level risk estimation.

$$R_{\text{media}} = 0.40\, S_{\text{image}} + 0.35\, S_{\text{capt}} + 0.25\, S_{\text{hash}} \tag{1}$$

The weighting scheme (0.40, 0.35, 0.25) is determined based on the relative psychological and behavioral influence of each modality, assigning the highest contribution to visual content due to its stronger cognitive and emotional impact on minors. Caption-based textual content is given the second highest weight as it provides explicit semantic and affective cues, while hashtags receive a comparatively lower weight since they primarily offer contextual reinforcement rather than direct expressive information. To reflect developmental vulnerability and real-world usage behavior, the risk score obtained is obtained by summing up developmental and contextual modifiers with the base media score.

$$R = R_{\text{media}} \cdot A_f + G_a + C_m \tag{2}$$

The parameter A_f denotes the age-based adjustment factor, G_a represents the gender-based behavioral bias correction term, and C_m captures contextual influences such as late-night posting behavior. Children aged 10–13 receive a higher age factor to account for lower impulse control and higher susceptibility to addictive stimuli, whereas older adolescents receive lower adjustment values reflecting improved self-regulation. The final score is normalized to a 0–100 range and classified as Low (0–30), Medium (31–70), or High (71–100) risk. To improve transparency, a fine-tuned Fine-tuned Language Net - Text-To-Text Transfer Transformer (FLAN-T5) transformer generates plain-language explanations for each assessed post, enhancing interpretability for parents and guardians.

B. Multimodal Complaint-Driven Risk Analysis

This module analyzes structured and unstructured complaints submitted by parents or guardians through the SafeKid Scan web interface. Complaints may be provided in text or voice form; all speech inputs are first transcribed into text. Each submission includes contextual metadata such as the child's age, gender, region, average daily social media usage, primary device type, and a detailed complaint description. Data are collected from verified sources, including child protection agencies and mental health hospitals. Textual data are preprocessed to enhance quality and consistency by eliminating noise, redundant terms, and emojis, subsequently undergoing case normalization, whitespace standardization, and orthographic correction The processed text is analyzed using a supervised transformer-based classifier, referred to as MentalRoBERTa. This RoBERTa-based model is fine-tuned on mental health and behavioral datasets and is effective in identifying addiction-related linguistic and emotional patterns.

The classifier produces a risk score in the range of 0–100, representing the severity inferred from the complaint content. Experimental evaluation on institutional datasets reports approximately 85.0% training accuracy and 85.6% validation accuracy. In addition to the learning-based analysis, a rule-based scoring system evaluates at clinically relevant behavioral signs such as frequent screen usage, trouble sleeping, falling behind in school, feeling sad, and withdrawing from social situations. The total complaint-based risk score is obtained by a combination of learning-based, or rule-based, and past indicators.

$$R_{\text{complaint}} = \alpha \cdot S_{\text{ML}} + \beta \cdot S_{\text{RB}} + \gamma \cdot S_{\text{H}} \tag{3}$$

The variables S_{ML}, S_{RB}, and S_{H} represent the machine learning–based, rule-based, and historical trend scores, respectively. The coefficients α, β, and γ are experimentally determined. The resulting score is normalized to a range of 0–100 and categorized into low-risk, medium-risk, and high-risk levels. The module produces a synthesized report summarizing the identified behavioral indicators, associated risk level, and temporal patterns. The combined formulation allows the system to combine the short-term indicators of behavior and long-term use conditions, making sure to have an adequate and sound evaluation of the risk of addiction. A combination of historical trends and machine learning and rule-based indicators will make the model less sensitive to short-term fluctuations, and more reliable in predicting risks. Created structured report allows making informed decisions by caregivers and professionals as it will justify the assigned risk level clearly and help to plan intervention and follow-up in time.

C. Automated Risk Evaluation and Blockchain-Based Report Integrity Framework

In the quest to provide credible, ethical, and explainable decision-making in child social media addiction assessment, a Risk Score Conflict Detection and Resolution Mechanism is proposed in the proposed system. The current module has been envisioned to detect and balance inconsistencies between risk scores that were independently generated out of different behavioral views. Media Analysis Risk Score (R_{m}):Derived from the child's digital behavior, including social media usage duration, content interaction patterns, and temporal activity characteristics. Complaint Analysis Risk Score (R_{c}):Derived from real-world observations reported by parents or authorities, such as behavioral changes, emotional instability, sleep disruption, and academic impact. Since these scores originate from independent data sources, direct consistency between them cannot be assumed. To measure the degree of agreement between the two analytical perspectives, the system determines a Risk Gap (RG) through deterministic formulation.

$$R_G = |R_{\mathrm{m}} - R_{\mathrm{c}}| \tag{4}$$

The absolute difference allows making a clear comparison between risk assessments. The categories of compliance are compliant (0-15), partially non-compliant (16-40) and highly non-compliant (40-60). In the analysis, the system will result in a final, AI-generated report outlining the addictive condition, instructions on how to remove the content, and the necessary legal or professional actions. To make sure the report is tamper resistant, the report is hashed using SHA-256 and only the hash and a small amount of metadata are stored on a permissioned blockchain, which ensures that the report is immutable, verifiable, and its metadata is privacy preserved. Under the internal processing pipeline, PDF-uploaded documents are first unzipped and handled through PyPDF2. The content extracted is then purified and analyzed with regular-expression techniques and FLAN-T5 model and then validated before it is saved in the database.

D. IoT-integrated behavioral monitoring with digital health cards

Continuous behavioral monitoring based on IoT and the subsequent generation of alerts and treatment are offered in this module. The purpose of this methodology is to detect abnormal behavioral patterns early and be able to intervene in time with the assistance of an integrated digital health card. Wearable IoT sensors, heart rate sensors, body temperature sensors, motion sensors, Electrocardiogram (ECG) signal sensors, and microphone sensors, are used to gather physiological and behavioral data. These sensors monitor the information about stress, agitation, restlessness, emotional instability, and abnormal activity patterns. The obtained information is taken in form of time-series signal and transmitted to the backend system at periodic intervals. The data obtained is divided into windows of a fixed length. Sensor streams are analyzed to obtain time-domain features, including mean heart rate, heart rate variability, motion intensity and temperature changes. The values of all features are standardized through the normalization methods in order to accommodate mixed sources of sensor.

The characteristics are classified by a Random Forest classifier in order to distinguish between normal and stress conditions which are induced by addiction. The classifier selected is the Random Forest as it is resistant to noisy data produced by IoT devices and also because it works better with mixed feature sets. The training accuracy and the validation accuracy of the experiment were 96.88% and 96.97% respectively. Follow-up reminders are also automatically activated in order to ensure that recovery plans are adhered to. The intensity of the reminders is adjusted dynamically according to the risk levels, previous behavioral pattern as well as the success of previous interventions.

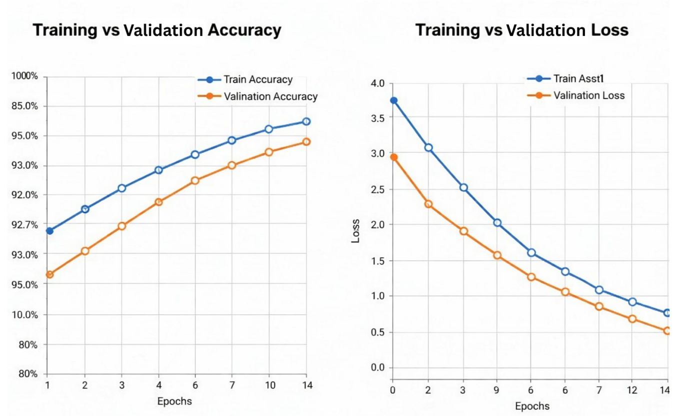

Fig. 2: Training and Validation Accuracy of the Image Analysis Module

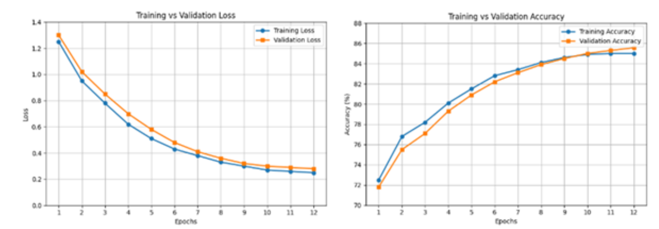

Fig. 3: Training and validation accuracy of the Complaint Analysis Module

IV. RESULTS AND DISCUSSION

To analyze the work of the proposed SafeKid Scan framework, a varied range of data occupying the social media content, behavioral complaints, and physiological sensor signals was used as the input. The aim of the test was to test the reliability, accuracy, and robustness of the multimodal risk detection pipeline in real-world operating conditions. All datasets were normalized before training so that they could be consistent and cause a modality-specific bias to be minimized. When tested on about 34,000 annotated instances of social media, the multimodal media analysis module had high capability to predict.

As demonstrated in Figure 2, the model followed a consistent convergence curve with a training accuracy of 95.88% and a validation accuracy of 95.97%. When training and validation curves are very close, the model exhibits minimal overfitting and effective generalization on unseen data, indicating stable and robust performance across all tested scenarios. Besides accuracy, the model achieved a precision of 95.12%, recall of 95.43%, and F1-score of 95.27%, demonstrating reliable identification of normal versus addiction-related content. The visual features with caption and hashtag semantics greatly increased detection performance in comparison to single-modality solutions, which serves to demonstrate the relevance of multimodal representations learning to behavioral risk assessment. The behavioral risk analysis module based on complaint was tested on 2,400 complaint records grouped on experts who were provided with complaint records in English and Sinhalese.

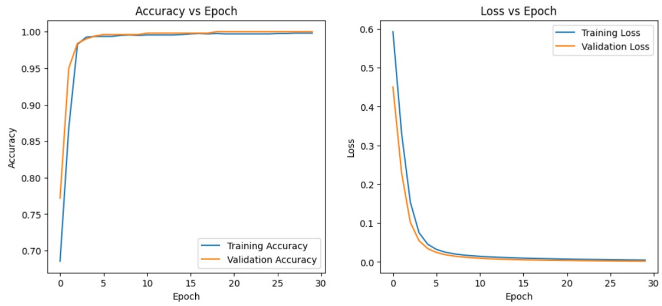

Fig. 4: Stable accuracy convergence in the IoT module

The transformer-based classifier has both a training accuracy and a validation accuracy of 85.0% and 85.56%, respectively as indicated in Figure 3. These findings reflect the strength of this model in terms of addressing linguistic variability and casual narrative patterns that are typical of parental and guardian complaints. High recall rates were noted in high-risk classes and this is very important in child protection systems where the minimization of false misses is a major consideration. Inference approach based on machine-learning was combined with rule-based behavioral indicators, which made the risk scores generated by this approach even more interpretable and reliable. IoT-based activity and physiological monitoring served as an additional source for verifying behavioral risks. The IoT module was trained and validated using 2,500 sensor records, achieving a training accuracy of 96.88% and a validation accuracy of 96.97%. The performance of this module is illustrated in Figure 4. Such findings prove the efficiency of physiological variables like heart rate variability and motion intensity in determining stress patterns related to excessive digital activity. The convergence stability is high and this means that it is not sensitive to sensor noise and time changes.

Lastly, multimodal media analysis, complaint-based behavioral assessment, and IoT-derived physiological insights were incorporated, which enabled an end-to-end assessment of the digital addiction risk in minors. The proposed framework integrates heterogeneous sources of data and makes it less dependent on a single modality and more resistant to noisy or incomplete data. The trust, transparency, and accountability of the automated report-generation mechanism are enhanced with the help of blockchain-based integrity checking because risk-assessment results are immutable and verifiable. Moreover, integrateable AI elements allow parents, guardians, and professionals to understand the elements that lead to the assigned risk levels, which in turn helps them to make informed decisions and implement intervention strategies on time. It has been shown that experimental results are highly accurate, stable, and generalised with a wide range of datasets and thus substantiates the usefulness of the given architecture in the real world. In general, the conclusions described above demonstrate that SafeKid Scan is a valid, scalable, and privacy-conscious tool to detect and manage the risk of digital addiction early on, which offers a feasible basis to apply in child-protecting and mental-health monitoring systems.

V. Conclusion and Future Work

This paper presents the cross-modal behavioral risk monitoring system, SafeKid Scan, designed to detect early signs of digital addiction in minors. The framework integrates multimodal social-media analytics, complaint-based assessment, AI-driven risk scoring, blockchain-secured report integrity, and IoT-based behavioral monitoring. Although the system shows strong predictive performance and cross-modal generalization on heterogeneous data, it has limitations related to dataset dependency, sensor variability, and the need for large-scale real-world validation. Ethical aspects such as privacy preservation, informed guardian supervision, and secure data handling were considered throughout development and evaluation. Future work will focus on improving scalability, optimizing multimodal model efficiency, and extending coverage to diverse languages, regions, and sensor modalities, while refining risk-scoring models using larger datasets for reliable real-time deployment.

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